L O T   S Y S T E M S   C O R P O R A T I O N

LOT PAGER

HARDWARE MANUAL — v1

"One button. One line of text.
One gesture to record it."

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brand.lot-systems.com

01 — WHAT THE PAGER IS

LOT Pager is a physical, silent, one-line notification object. It receives ambient text from the LOT Memory Engine ("Coffee time.") and shows it on a small screen. No app, no scroll, no push alert — the screen updates silently and only shows the newest line when glanced at.

FORM FACTOR

- 40mm x 40mm x 5mm flat silver square puck, two stainless steel shells
- BACK: mirror-polished stainless steel, no seams, no visible ports
- FRONT: brushed stainless steel — camera aperture, display window, COPY button
- 5mm height is an aggressive target; 6-8mm is the engineering fallback
- Qi charging through steel requires a non-metal window over the coil

CORE ELECTRONICS

- MCU: ESP32-S3-WROOM-1 (WiFi + BLE, camera bus, deep sleep)
- Display: 0.96" monochrome OLED, SPI, single line of text
- Camera: Arducam OV2640 Mini, 2MP — QR/pairing scan + presence capture
- Weather: Bosch BME280 (temp / humidity / pressure)
- Charging: Qi receiver (BQ51013B-class) + external charging puck
- Battery: 3.7V ~150mAh LiPo pouch cell, 3.8mm thick

02 — BILL OF MATERIALS (SUMMARY)

Full links + sourcing detail in LOT-PAGER-BOM.md. Figures below are rough order of magnitude at 100-unit volume; PCB and enclosure lines require a real PCBWay quote before budget lock.

COMPONENT	x100 EST / UNIT
MCU — ESP32-S3-WROOM-1-N8	$3.50 - 4.00
Display — 0.96" mono OLED (SPI)	$3.00 - 5.00 (QUOTE)
Camera — Arducam OV2640 Mini	$8.00 - 12.00 (QUOTE)
Weather sensor — Bosch BME280	$1.50 - 2.50
Qi receiver (IC + coil)	$2.50 - 4.00 (QUOTE)
Button — SMD tactile, low profile	$0.30 - 0.40
Battery — LiPo 150mAh pouch	$2.00 - 3.00 (QUOTE)
PCB fab + SMT assembly (PCBWay)	QUOTE
CNC stainless shells x2 (PCBWay)	QUOTE

SENSOR GRADE INDEX (SGI) — AI-ASSISTED SELECTION

Every sensor/camera/charging candidate is scored 0-100 across accuracy, power, footprint, and availability before lock — full scoring table in LOT-PAGER-BOM.md section 05.

03 — ROADMAP

PHASE 0	Weeks 1-2	De-risk 5mm z-height; resolve Qi-through-steel window decision
PHASE 1	Weeks 3-6	KiCad layout; 5-piece PCBWay prototype run; firmware bring-up
PHASE 2	Weeks 7-8	Backend: device pairing + token auth, notification transport
PHASE 3	Weeks 9-10	CNC stainless prototype x5; fit-check; Qi coil placement
PHASE 4	Weeks 11-12	5 hand-built units; S-2 dogfood; fix loop
PHASE 5	Weeks 13-16	100-unit PCBA + CNC run; final assembly; PDF manuals; ship

OPEN RISKS

- 5mm total height — the hardest physical constraint in the spec
- Wireless charging through stainless steel needs a non-metal window
- No push/notification infrastructure exists server-side today — greenfield build
- No device-token auth path exists today — greenfield build
- RF (WiFi/BLE) range inside a metal enclosure needs its own antenna check

Companion documents: LOT-PAGER-SPEC.md, LOT-PAGER-BOM.md, LOT-PAGER-FIRMWARE.md, LOT-PAGER-API-CONNECTOR.md

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